Arboriculture & Urban Forestry 2024. 50(5):346–364
https://doi.org/10.48044/jauf.2024.018

ARBORICULTURE & URBAN FORESTRY
Scientific Journal of the International Society of Arboriculture

Why Can't We All Just Get Along? Conflict and Collaboration in Urban Forest Management

By Stephanie Cadaval, Mysha Clarke, Lillian Dinkins, Ryan W. Klein, John W. Roberts, and Qingyu Yang

Abstract. Urban forest management is crucial for supporting human well-being, ecosystems, and society, particularly with expanding global urban population and multi-uses of these urban greenspaces. This literature review examines the conceptualization and factors that contribute to conflicts and/or collaborations in urban forest management, including, but not limited to, diverse actors' uses, needs, and perceptions. Using PRISMA methods, we systematically reviewed 176 scholarly articles published between 2013 and 2021 and found that most articles were primarily from the United States, Australia, and Canada. Findings highlight the need for clearer definitions of collaboration, emphasizing communication, operational tasks, planning, and shared beliefs among actors. Positive collaborations involved multi-level engagement and inclusive decision-making. In most cases, multiple issues contributed to conflict, including a variety of stakeholders with differing viewpoints on a given situation. Conflicts are commonly complex situations that do not lend themselves to a one-size-fits-all solution and tend to be a unique manifestation of the people, places, and perspectives involved. Our review can inform practitioners about more inclusive practices and adaptive management of urban forests. We conclude by providing lessons learned and suggestions for future research on stakeholder involvement, public education, governance, policy, decision-making, and the role of biophysical and ecosystem services in urban forest collaboration and conflicts.

Keywords. Stakeholder Engagement; Urban Forest Governance; Urban Forest Management.

INTRODUCTION

Urban forests provide a plethora of ecosystem services (Davies et al. 2018;Bonilla-Duarte et al. 2021). Urban forests also provide a critical space for nature interactions, as the majority of the world's population lives in urban areas (United Nations 2018). Therefore, the socio-ecological nature of urban forest management engages with multiple actors across temporal and geographic scales. These diverse actors' values and perceptions can incite conflicts or opportunities to collaborate on urban tree management, governance (Kirkpatrick et al. 2013), and the provisioning of ecosystem services and disservices (Kirkpatrick et al. 2013; Roman et al. 2021). Governance is the act of coordinating action towards accomplishing goals (Kjær 2004). Specifically, urban forest governance involves complex decision making and diverse actors. Actors are those individuals and collective groups that influence or make decisions about urban forests or individual urban trees (Lawrence et al. 2013). Actors can also influence urban forest decision making based on the resources and discourses they engage in and their collective engagements (Ordóñez et al. 2023). Although local authorities are important, urban forest governance also incorporates the values, priorities, and management pressures from the public and private sectors and civil society (Konijnendijk van den Bosch 2015).

Collaboration is recognized as a pivotal component of urban forest management (Doucet et al. 2024), but there are only a few studies that investigated the best strategies for collaboration (Vasquez-Brust et al. 2020). Governance that facilitates collaboration and reduces conflict is increasingly important as most of the world's population lives in urban areas (United Nations 2018). Governance that facilitates collaboration has shared goals, relationships of trust, and information sharing (Lopes et al. 2020; Wirtz et al. 2021).

Effective governance approaches, particularly in urban forestry, include financial resources, decisions driven by data, and setting goals, objectives, targets, and shared visions (Wirtz et al. 2021).

Human behaviors and decision making influence urban forest function through planning processes, tree species selections (Hilbert et al. 2023), maintenance, and removals (McPherson et al. 2003). Owing to the diverse uses, perceptions, and preferences surrounding urban forest management, conflicts can occur among actors, including (1) differing values between professionals such as municipal managers, developers, and arborists (Kirkpatrick et al. 2013; Kronenberg 2015; Ordóñez et al. 2019); and (2) differing priorities between forestry professionals and the public (Carmichael and McDonough 2019). For example, city planners and arborists may have contrasting perspectives about where to place trees and how to maintain them in the urban landscape (Kirkpatrick et al. 2013). Likewise, urban residents and forestry professionals may disagree about tree risk perceptions (Clark et al. 2020). Conflicts around urban trees and associated divisive values can foster a lack of community engagement and challenge stewardship initiatives (Kronenberg 2015). Likewise, conflicts can occur at diverse levels and scales of human-nature interactions. Urban forest management conflicts can involve (1) contrasting priorities between citizens and non-profit tree advocacy groups (Carmichael and McDonough 2019); (2) coordination challenges between municipal forest managers and urban development actors; and (3) general risk aversion attitudes of the public (Ordóñez et al. 2020b).

On the other hand, collaboration may be equally varied in urban forest management and may occur between the public and municipal tree managers and between local governments and civic organizations. Collaboration between the public and professionals can motivate stewardship and engagement (Nesbitt et al. 2019) and provide a way to overcome challenges faced by natural resource managers with limited resources and capacity (Nannini et al. 1998). For example, residents who have previously planted trees shared a willingness to partner with municipal urban forestry initiatives (Conway and Bang 2014). In a case study analysis of Boston (Massachusetts), Baltimore (Maryland), and Philadelphia (Pennsylvania), effective partnerships were established between the cities' Parks and Recreation departments and non-profit organizations that support urban tree initiatives (Foo 2018).

To our knowledge, there is no comprehensive literature review on an international scale about conflict and/or collaborations related to urban forest use, governance, perceptions, and management. Previous research on urban forestry studied challenges and successes of collaboration between non-government organizations and local governments in 9 Canadian cities (Doucet et al. 2024), Latin America (Ordóñez et al. 2023), and the Caribbean (Devisscher et al. 2022). Previous literature reviews studied urban forest managers' perspectives on governance and decision making (Ordóñez et al. 2019), the link between urban forest use, management, and social problems (Referowska-Chodak 2019), and the types of conflicts experienced in densifying greenspaces (Madureira and Monteiro 2021). Previous studies highlight that a key cause of conflicts in urban forest management include disputes between recreational users (Wilkes-Allemann et al. 2015b). Furthermore, there is a spectrum of perceptions about urban trees which may lead to conflicts because of the varied values and attitudes held about trees and depending on the type of forestry professional involved (Kirkpatrick et al. 2013). Collaboration presents an opportunity to include all interests and a shared vision (Elmendorf and Luloff 2001). Having shared aims and goals can facilitate collaboration in urban forest management. Shared goals might include securing natural spaces in and around urban environments (Ruliffson et al. 2002), collaborating around environmental stewardship through outreach, research, and sharing knowledge (Nannini et al. 1998; Driscoll et al. 2012), and developing opportunities for social learning between the public and government (Yamaki 2016). Collaboration among urban forestry professionals also creates opportunities for sharing resources and building relationships (Cadaval et al. 2024).

The purpose of this review is to assess: (1) how conflict and collaboration have been defined in previous studies about urban forests, including relevant socio-ecological contexts; (2) the actors involved in urban forest conflicts and collaborations; (3) lessons learned about conflict and collaboration in urban forest management; and (4) opportunities for future research on conflict and/or collaboration in urban forest management.

MATERIALS AND METHODS

We searched ScienceDirect, SCOPUS, and Web of Science databases to find peer-reviewed journal articles, literature reviews, and perspective articles, excluding book chapters and editorials. The search was open to any time period and geographic location. We included literature in English and Spanish languages. We identified relevant articles using: (1) "urban" OR "cities" OR "metropolitan" OR "towns" OR "municipal" AND (2) "trees" OR "forests" AND (3) "collaboration" OR "cooperation" OR "partnerships" OR "coordination" OR "collabo*" OR "partner*" OR "conflict" OR "disagree" OR "dispute." We used the PRISMA flow diagram to systematically document the search results (Figure 1)(Moher et al. 2009). The eligibility criteria for articles included studies that focused on one or multiple urban areas, research that focused on conflict and collaboration in urban forest management at the policy and/or stakeholder level, and excluded studies that did not specify an urban area.

The initial search returned 4,409 articles from SCOPUS (2,284), Web of Science (1,537), and ProQuest (588). These articles were then imported into the Mendeley bibliography management software. After deleting 2,631 duplicate records, 1,778 were imported into Microsoft Excel to be screened by 4 reviewers against eligibility and exclusion criteria. At this phase, each reviewer screened the title and abstract and retrieved 345 articles for full-text review (Figure 1).

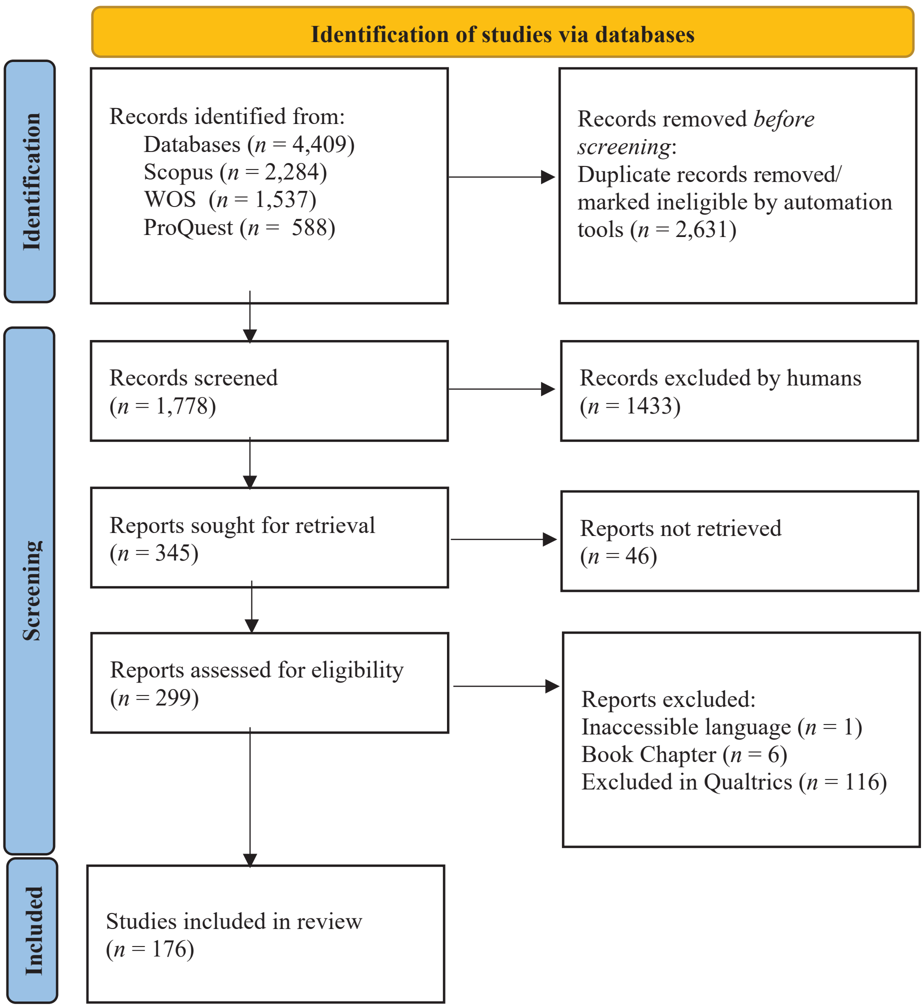

Figure 1. PRISMA flow diagram of the resource-screening and article-selection process. Adapted from Page et al. 2021.

The final remaining articles were thoroughly analyzed qualitatively and quantitatively using codes deducted from our research questions (Table 1). To minimize bias, we conducted an intercoder reliability test where a selected reading was shared among the research team and separately coded by each. The final coding process was completed using a mix of QSR NVivo (Lumivero 2020) and a Qualtrics online form available through the University of Florida. Using a Qualtrics online form allowed multiple reviewers to code different articles and provided a way to organize the results. On the other hand, other review questions were coded qualitatively to get more in-depth responses such as lessons learned, definitions of conflict or collaboration, and opportunities for future research (Table 1). Following the PRISMA procedures (Figure 1), the final number of included articles for review was 176.

RESULTS

Description of Literature

From the 1,778 screened articles, we included 176 papers for final analysis, the majority of which are research papers (134), reviews (13), opinions (6), and others, such as case studies (23). All articles were published between 1989 and 2021, although most of them were published in the last decade (Figure 2) in primarily *Urban Forestry & Urban Greening* (40), *Forests* (9), *Landscape and Urban Planning* (8), *Journal of Arboriculture* (7), *Arboriculture & Urban Forestry* (6), *Cities* (3), and *Forestry* (2). However, some papers (97) were published in journals specializing in other disciplines.

Out of the 176 scholarly articles, 63 used qualitative methods (i.e., interviews, focus groups, participant observation, document analysis, etc.), 56 used quantitative

Table 1. Codes stemmed from the research questions and study goals and the expected responses of each.

Review questions	Possible responses
What are the definitions of conflicts in urban forest management provided in the paper?	Open-ended and coded qualitatively
What are the definitions of collaborations in urban forest management provided in the paper?	Open-ended and coded qualitatively
What research methods were used?	Qualitative (interviews, focus groups, participant observation, etc.) Quantitative (US Census, survey questionnaire, etc.) Mixed methods Other, explain
What locations were studied?	Country, city/community block, etc.
What is the context around the conflicts and/or collaborations studied?	Open-ended based on research item and coded qualitatively
What types of theories, if any, were used to study conflict and collaboration?	Open-ended based on research item
What stakeholders were involved at what different scales?	Open- and close-ended based on research item using the following options: • Government agencies • Nonprofit or volunteer organizations • Extension agents • Private consultants/private forestry professionals • Non-expert public • Stakeholders within agency, between agency, between public/agency • Did not specify any stakeholders
What lessons were learned that can enhance the management of urban forests?	Open-ended based on research item and coded qualitatively
What are the research gaps/opportunities for future research on conflict and collaboration?	Open-ended based on research item and coded qualitatively

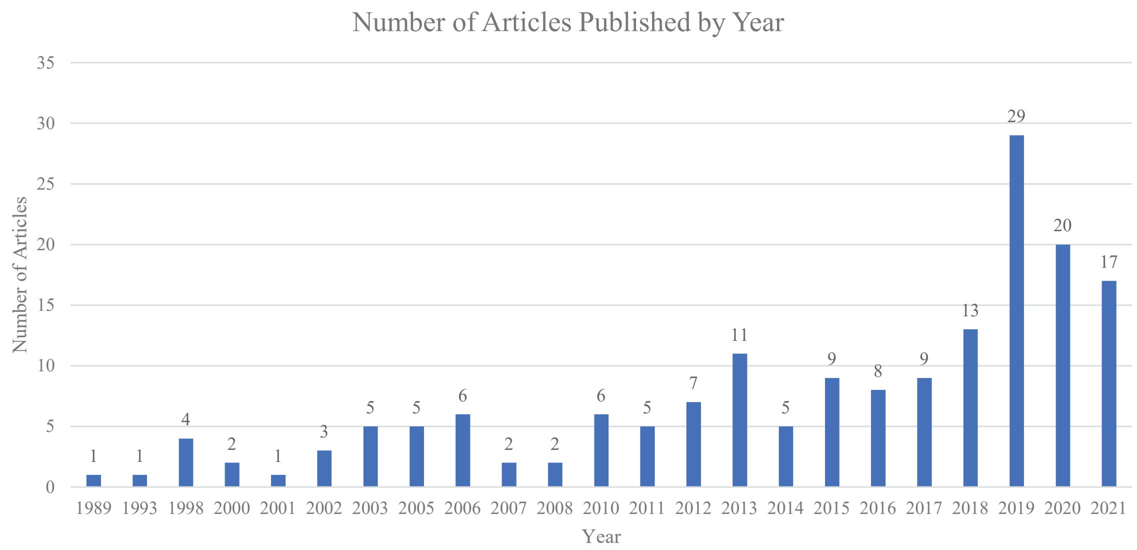

Figure 2. Number of articles published over time about conflict and collaboration in urban forestry.

methods (i.e., surveys, census data, or questionnaires), and 40 used a mixed methods approach, including a combination of qualitative, quantitative, and spatial, while 12 articles used spatial data only. The top 5 countries producing articles about conflict and collaboration in urban forests were the United States (57), Australia (10), Canada (9), the United Kingdom of Great Britain and Northern Ireland (8), and Germany (6)(Figure 3). Most articles focused on urban forestry issues at the city (81) and national levels (33), while the other papers addressed multi-national, regional, or smaller scales (54), like communities, neighborhoods, or college and university campuses (Table 2).

Table 2. Research scale used in literature about conflict and collaboration in urban forestry.

Research scale	Count
Community	12
City	81
Regional	30
National	33
Multinational	10
Other	54

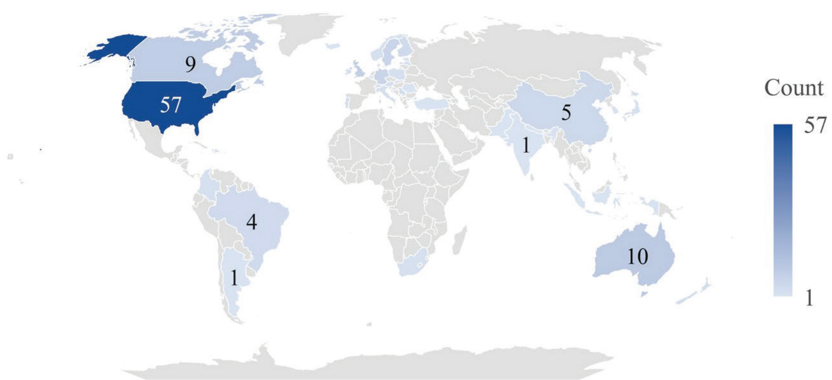

Figure 3. Geographic distribution of literature on conflict and collaboration in urban forestry. Darker shades illustrate where in the world more research is done that discusses conflict and collaboration in urban forestry. Count reflects how many articles were identified from each geographic region.

Frameworks and Theories Used

Of all the papers reviewed, 73 articles explicitly used theories, models, or frameworks to guide their research on conflict and/or collaboration in urban forests. The theories and frameworks were organized into 5 broad categories, including: (1) theories of motivation, cognition, and reasoning; (2) morals, intuitions, culture, and identity-based theories; (3) trust, negotiations, and public involvement; (4) organizational theories; or (5) systems theories. Most articles that used a theory/framework focused on trust and public involvement or systems theories (Table 3).

Collaboration and Conflict in Urban Forest Governance and Management

Our literature review revealed various contexts, issues, and priorities regarding urban forest management, governance, and use that impacted collaboration and/or conflict. These discussions about conflict and collaboration revolved around the intricacies of urban forestry projects (Sipilä and Tyrväinen 2005; James et al. 2009; Young 2011), governance (Pinto et al. 2016; Kozová et al. 2018; van der Jagt and Lawrence 2018; Ordóñez et al. 2020a; Frantzeskaki and Bush 2021; Pinkerton and Rutherford 2021), or social concepts like values and attitudes (Tyrväinen et al. 2003; O'Brien 2005; Muller and Bornstein 2010; Stobbart and Johnston 2012). However, precise definitions of these concepts were absent from most of the literature reviewed.

Understanding Conflict

Overall, our results found that conflicts in urban forestry governance resulted in setbacks, unforeseen obstacles, mismanagement, and disengagement with relevant stakeholders. In addition, conflicts also delayed program initiatives, creating a need to reevaluate management priorities and activities and, in some cases, leading to certain management efforts being completely forgone. In the literature, the causes of such conflicts varied based on the diversity of people involved, geographic location, time, and constraints, such as legal, ethical, financial, and public health and safety. Overall, conflicts within urban forestry did not conform to a one-size-fits-all model. Rather, there were many potential causes.

Our results show that the top contributors to conflicts in urban forest management included: unaligned or differing goals (96 articles) and lack of or poor communication with the public (61)(Table 4). We found several smaller categories describing the factors contributing to conflict as planning and policy or political issues (20), management (16), land use and development (15), infrastructure and resource allocation

Table 3. Theories used in articles describing conflict and collaboration.

Categorization	Theories	Number of articles (*n* = 73)
Theories of motivation, cognition, and reasoning	Grounded theory (1)	1
Morals, intuitions, culture, and identity-based theories	Framing (2); place attachment (2); Pradya's framework for inclusive design (1); Critical property theory/Postcolonial theory (5); Culture theory (2)	12
Trust, negotiations, and public involvement	Citizen science (3); stakeholder engagement (3); social impact model (3); participatory planning (3); interdisciplinary collaboration (3); collective memory/social learning (3); social diffusion (5); Janis and Mann Conflict Model of Decision-Making (1)	24
Organizational theories	Institutional analysis and development (4)	4
Systems theories	Ecosystem services (5); social-ecological systems (5); systems thinking/collaborative conceptual modeling (5); Clark's model of urban forest sustainability (5); policy arrangement approach (1); urban political ecology (1)	22
Other	Nature-based solutions (5); social housing regulatory framework (1); City Biodiversity Index (1); Liberal model of social integration (3)	10

(15), environmental concerns (11), climate change and sustainability (8), risk and disasters (5), and tree services and tree equity (10)(Table 4). Specifically, 37% of articles reported conflicts between professionals (Clark and Matheny 1998; Kuhns et al. 2002; Shapira et al. 2013; Pinkerton and Rutherford 2021), 62% reported conflicts between professionals and the general public (Ruliffson et al. 2002; Reams et al. 2005; Jankovska et al. 2010; McMillen et al. 2019), and 36% reported conflicts within the general public (Mattila et al. 2015; Wilkes-Allemann et al. 2015b; Kraxner et al. 2016)(Table 5). Lack of or poor communication was frequently cited as a cause for conflicts in urban forest management. These included conflicts that resulted from lack of or poor communication with other organizations (Wu et al. 2019; Madureira and Monteiro 2021), the public (Elmendorf and Luloff 2001; van der Jagt and Lawrence 2018), or internal conflicts within the institution (Pincetl 2010; Harper et al. 2018; McGrath et al. 2021). Lack of resources (Nannini et al. 1998; O'Callaghan and Skelly 2013), safety (Gasson and Cutler 1998; Conway and Bang 2014), unaligned or differing goals (Shapira et al. 2013; Davies et al. 2018), and maintenance issues also occasionally led to conflict (Breuste 2013).

Table 4. Factors that contribute to conflicts in urban forest governance.

What factors contribute to conflicts?	
Factors that contribute to conflicts	**Number of articles**
Unaligned or differing goals (i.e., management, development, land use, recreation, fire prevention, landscape preferences, invasive species management, etc.)	96
Lack of or poor communication with the public	61
Maintenance	54
Stakeholders and social factors	51
Lack of resources (i.e., budget and staffing, financial, environmental, access and inclusion, policy and planning)	47
Lack of or poor communication with other institutions	37
Safety	35
Lack of or poor communication within the institution	26
Planning, policy, or political issues	20
Management	16
Land use and development	15
Infrastructure and resource allocation	15
Environmental concerns	11
Tree services and tree equity	10
Policy and planning	9
Climate change and sustainability	8
Risk and disasters	5

Table 5. Overview of general categories for different types of conflicts.

What type of conflicts were identified?	
Frequency of conflicts (*n* = 143)	**Total**
Conflicts between professionals and the general public	89
Conflicts between professionals	53
Conflicts within the general public	51

Furthermore, most articles presented multiple layers of conflicts. For example, 66 studies with defined conflicts highlighted that such issues often feature multiple overlapping variables (e.g., management, policy and planning, budget and staffing, and public and private stakeholders)(Muller and Bornstein 2010; Nesbitt et al. 2019; Köse et al. 2021). Similarly, 94 articles without clear definitions of conflict also featured multiple overlapping variables, which provided context to the conflict such as conservation, development, social characteristics, and political characteristics (Zinzani and Curzi 2020; Rodriguez et al. 2021). Furthermore, there was considerable overlap across studies, wherein 30% of studies described multiple categories of conflict. Examples include studies that expressed both internal conflicts between professionals along with external conflicts between professionals and the public (Nordström et al. 2010; Morgenroth et al. 2015; Ordóñez et al. 2020a), as well as conflicts within the general public and conflicts between professionals and the general public (O'Brien 2005; Aasetre et al. 2016; Svendsen et al. 2021).

A more detailed look at unaligned or differing goals as a source of conflict within urban forestry showed that management (13), development (7), and land use (6) were the highest-ranking factors that contributed to conflicts. Conflicts about lack of resources were primarily attributed to budget and staffing (12), financial costs (12), and environmental concerns (3).

We also found that management (16)(e.g., Gundersen et al. 2006; Gutsch et al. 2019), land use (9)(e.g., Ballantyne et al. 2014; Pinto et al. 2016; Steenberg et al. 2019), and policy and planning (9)(e.g., Lawrence and Ambrose-Oji 2015; Brown et al. 2018) were frequently cited. Fifty-one (51) articles described conflicts centered on stakeholders and social factors.

Understanding Collaboration in Urban Forest Management

Definitions of Collaboration

Few papers specifically defined "collaboration" as a concept or within the context of a specific case or research question. Four research articles provided explicit definitions of collaboration as (1) achieving consensus (Shapira et al. 2013); (2) having "compatibilities" (Madureira and Monteiro 2021); (3) achieving interdependence and interaction by working through differences to find solutions while incorporating trust and shared visions (Elmendorf and Luloff 2001); or (4) when two parties have a vested interest in a public-private partnership (Nannini et al. 1998). In one perspective paper, collaboration meant sharing power to define problems and develop solutions by deconstructing barriers for participants (McGrath et al. 2021). On the other hand, 7 papers specifically discussed "partnerships" and their importance for effective governance (Kozová et al. 2018; Garrison 2019). Furthermore, developing and maintaining partnerships helped participants overcome challenges by relying on each other's experiences (Bierzychudek 2020; Svendsen et al. 2021).

Who is Included in Collaborative Urban Forest Management?

Globally, partnerships supporting urban forestry management included: collaborations between residents, community-based organizations, public-private collaborations, and partnerships between municipalities and vulnerable groups (Kuchelmeister 2000; Ryu and Chi 2021). For example, effective partnerships in urban forestry initiatives like tree planting in New York City included community groups, non-profit organizations, and government agencies (Moskell and Allred 2013). Actors involved in collaborative management included: (1) policy and management actors at local to federal levels; (2) extension agents and researchers; (3) actors in the private sector; (4) professionals in urban forestry fields; (5) volunteers, non-profit actors, master gardeners, and other related interest groups: and (6) "residents" or the "public."

Collaborations in urban forest governance happened at varying levels, including horizontally between legislative and executive institutions and vertically between institutions of land-use planning (Maruna et al. 2019). Non-profit organizations were also formed organically to support actors with diverse views about urban forest management actions, such as removing suburban recreational forests in Cape Town, South Africa (Day and Patel 2021). Through collaborative processes, communities and community-level actors are empowered to impact management actions, as exemplified in the development of a community forest and urban park in Migdal HaEmek, Israel (Gasul and Shmueli 2016), and the involvement of residents in monitoring to manage Dutch elm disease in California, USA (Nannini et al. 1998). Additionally, in the Midwestern and Eastern United States, collaborative tree planting efforts that included residents experienced increased neighborhood ties and positive resident perceptions about neighborhood beauty and place appreciation (Watkins et al. 2018).

Our review revealed other methods to engage the community in collaborative urban forest management, such as: public trainings and inclusion in tree-monitoring efforts (Nannini et al. 1998), shared planting and maintenance of city trees (McGrath et al. 2021), knowledge co-production and information-sharing (McGrath et al. 2021), community events and grassroots initiatives (Young 2011; De Smet and Van Reusel 2018), integrating collaborative decision making (McMillen et al. 2019), and designing collaborative structures into management programs and recruiting and empowering community actors to influence urban forest management activities (Gasul and Shmueli 2016).

Factors that Contribute to Collaboration in Urban Forest Management

Most articles described aspects of collaboration, including communication, process, and shared beliefs and desires (Table 6). Most papers (67) discussed communication with the public as central to collaboration and shared goals. Communication with the public about tree planting initiatives in community greening programs varied and sometimes resulted in resistance to partner with urban greening efforts, for example, when residents are not effectively consulted or supported (Carmichael and McDonough 2018). Likewise, research revealed a need to further engage with and inform the public about urban forestry and tree planting (Moskell and Allred 2013) to avoid disruptive interactions and improve the efficacy of urban forestry planning (Wilkes-Allemann et al. 2015a).

Sixty-five articles described other factors contributing to collaboration in urban forest management, including resources, inclusion, and leadership. Among the 12 papers that identified resources as a key component of collaboration, 3 types of resources were highlighted: people (Kuchelmeister 2000; O'Brien 2005; Brown et al. 2018), information and data (Gutsch et al. 2019; Alexander et al. 2020; Dupéy et al. 2021), and financial resources (Pincetl 2010; Dupéy et al. 2021). Various studies highlighted the importance of inclusion, accessibility (Ugolini et al. 2015; Nesbitt et al. 2019; Ordóñez et al. 2019), understanding (Nordström et al. 2010; Nguyen et al. 2017; Sotirov et al. 2017; Dupéy et al. 2021), and diversity (Elmendorf and Luloff 2001; Zinzani and Curzi 2020; Köse et al. 2021; Pinkerton and Rutherford 2021) in collaboration.

How Collaboration Is Categorized and Suggestions for Incorporating Collaborative Processes

Collaboration in urban forests was generally categorized as occurring (1) between professionals and the general public (70); (2) between professionals (97); or (3) other observations (102). Among the papers that discussed collaboration, several key themes describe how professionals engaged with the general public, such as (1) tree planting programs and public engagement; (2) urban forest management challenges, like public views about urban trees or management practices, risk aversion, and increased demand for green spaces; and (3) inclusive decision making and governance. Residents could be engaged in urban tree planting and retaining trees through bottom-up planning and place-making (De Smet and Van Reusel 2018) or through providing data and feedback that informed attitudes towards trees and preferred planting behaviors (Dilley and Wolf 2013).

Twenty-two papers studied some aspects of collaborative planning in urban forests and focused on resident or public inclusion or inter-institutional cooperation. Several papers highlighted contexts where resident participation was valued and incorporated, thereby leading to fewer conflicts throughout planning processes (Sipilä and Tyrväinen 2005; Stobbart and Johnston 2012; Dilley and Wolf 2013; De Smet and Reusel 2018). In some cases, goal setting with residents about urban forests and early inclusion in planning reduced conflicts (Sipilä and Tyrväinen 2005; Stobbart and Johnston 2012; De Smet and Reusel 2018). Additionally, understanding and collecting social data about residents informed the development

Table 6. Factors that contributed to collaborative processes in urban forest management.

Collaborative processes (larger themes)	Examples of collaboration	Number of articles referenced
Communication	Communication within organizations	30
	Communications with other organizations	52
	Communication with the public	67
Process or operational features	Maintenance schedules and defined tasks	12
	Meeting frequency and access	11
	Organization structure	20
	Use of strategic plan	45
Shared beliefs and desires	Shared goals	67
	Shared values	59

of public outreach programs at more local scales (Dilley and Wolf 2013). Specifically, understanding perspectives held by diverse actors informed the acceptability of urban trees, planting, and their diverse characteristics. For instance, older residents were more likely to view larger trees as having greater potential to cause property damage versus younger residents (Fernandes et al. 2019). Residents who lived on streets lined with certain species, like black poplar trees (*Populus nigra*), were less likely to “like” trees and associated trees with increased litter (Fernandes et al. 2019). Understanding local perspectives helped develop resources to inform residents’ acceptability of street trees and provided contexts for why the public may be reluctant to or challenge tree planting. Likewise, understanding the negative relationship between people and urban forests guided management strategies (Referowska-Chodak 2019), as trees occupied different meanings in various cultural spaces over the course of residents’ lives (Shimada and Johnston 2013). Trees that require little ongoing maintenance (i.e., conifers that minimize leaf clean up) and those that are smaller in stature, as well as low-cost planting material and subsidies for tree pruning also increased the number of willing participants or improved tree acceptance (Conway and Bang 2014).

Collaborative processes also occurred among institutional actors. Specifically, urban planning for city trees required greater inter-institutional coordination (Muller and Bornstein 2010; Bonilla-Duarte et al. 2021; Cheng et al. 2021), and conflicts could be reduced when interested parties felt included in the planning process (Wilkes-Allemann et al. 2015b). Neighboring municipalities that coordinated and shared governmental contracts reduced staff time, streamlined operations, cut expenses, shared data and information, and prevented competition for a limited number of available contractors at the municipal level (Alexander et al. 2020). Other research noted that urban forest management is enhanced by involving differing needs or perspectives based on sociodemographic factors and differences (Fraser and Kenney 2000; Jorgensen and Anthopoulou 2007).

DISCUSSION

Our systematic literature review provided insights about the factors that contributed to conflict and collaboration in urban forest management, including opportunities for better governance. Overall, there was little to no universal definition of conflict and/or collaboration used in the literature. The majority of papers published about conflict and collaboration in urban forests came from English-speaking countries, despite the global increase in urban populations and importance of urban forests for urban nature interactions. Additionally, the diversity of journals in which papers on urban forest conflict and collaboration are found suggested that research on this subject is widely relevant and interdisciplinary. Our research was limited to only articles on urban trees and forestry; future research could investigate conflicts and collaboration in other urban greenspaces or natural-resource contexts.

After examining the various contexts presented in these articles, we defined conflict and collaboration based on several conditions. The challenges that lead to conflicts included disagreement over goals, especially management, development, and land use. Additionally, conflicts stemmed from procedural and operational challenges, such as lack of resources, maintenance, safety, and communication within and with other groups or actors. Collaboration processes could support governance of urban forests by incorporating conflict-management actions. Through a cycle of interaction, developing trust, and interdependence among actors, participants could identify barriers, challenges, and differences, and begin processing conflicts and develop solutions. Feedback in the collaboration cycle strengthened the development of trust and encouraged continued engagement among actors.

Collaboration and conflict could occur in different phases of urban forest management, including defining management priorities (including the design/planning phase), management goals, maintenance activities (tree planting and removals, etc.), and how urban trees or forests were perceived and used. Each phase presented conditions and feedback that informed whether conflict or collaboration was present or possible, including a range of factors that contributed to collaborative/conflictive outcomes.

Urban forest managers believed urban forest management is successfully supported by including (1) sufficient financial resources; (2) data-informed decision making; and (3) clearly identified goals, objectives, and targets (Wirtz et al. 2021). However, urban forest management could be challenged by (1) lack of funds or insufficient resources; (2) lack of support from internal and external actors; (3) the pressure to develop urban areas; and (4) barriers to developing laws and policy (Wirtz et al. 2021). Through enacting collaborative processes, many challenges could be addressed, and actors could develop a shared sense of responsibility and leverage resources to ensure support for shared visions and goals (Wirtz et al. 2021).

The conflict and collaboration literature paints a complex picture of how people and trees interact in cities, highlighting the relationships that shape urban landscapes. Collaboration emerges as a critical element in responding to widespread urban forest challenges, such as invasive species (Alexander et al. 2020). For example, the Emerging Pests in Colorado (EPIC) working group and the associated Colorado EAB Response Team (CORT) acknowledges interagency collaboration as key to their success in preparing for the arrival of emerald ash borer (EAB), *Agrilus planipennis* Fairmaire (Coleoptera: Buprestidae). Preemptive response and education was expedited by organizational interactions rooted in professionalism and characterized by open communication and clearly defined agency roles, responsibilities, and limitations (Alexander et al. 2020). System-wide impacts could be addressed by involving multiple agencies and emphasizing resources, action, and knowledge sharing (Alexander et al. 2020).

The literature emphasized interconnected collaboration through community-engagement actions and inclusive decision making. To combat conflicts related to urban forest use, Wilkes-Allemann et al. (2015b) emphasized that communication played a pivotal role. Furthermore, response strategies should reflect local contexts and include all relevant actors. As beneficiaries to the services provided by urban trees, input by urban residents can be valuable to planners and managers, informing decision makers about use, preferences, and perspectives. Previous studies showed collaborative processes between citizen organizations and government agencies (Yamaki 2016) and the attitudes and perceptions of diverse actors toward professional collaboration opportunities and knowledge transfer (Ugolini et al. 2018). Collaboration with urban residents played a crucial role in shaping urban forestry programs. However, diverse actors may hold differing experiences, perceptions, and values about trees, and may lead to urban tree resistance.

The relevant challenges and opportunities for collaboration showed a need for more adaptive management and a cultural shift towards greater inclusion by managing ecological, social, and economic dimensions of urban forests. Collaboration is crucial for the sustainable and effective management of urban forests and trees and could enhance social learning and networks (Yamaki 2016). Conflicts among diverse actors could be reduced by including greater public participation in the planning process, increasing public education and engagement, and incorporating local dimensions and contexts into tree management (Referowska-Chodak 2019). Similar and overarching goals about urban forest management created opportunities to collaborate (Stobbart and Johnston 2012). Studies about coordination included various entities like government agencies and private actors (Pincetl et al. 2013). Tree planting initiatives where residents and non-profit organizations or municipal governments collaborated increased neighborhood ties among participating residents (Watkins et al. 2018).

The literature presents a need for mutual understanding and cooperation among different actors. Understanding cultural values and histories is crucial in shaping public attitudes towards urban forestry, emphasizing the need for culturally sensitive approaches in planning and management. Cultural backgrounds influence perceptions of urban trees, and the education and engagement of leaders is critical for developing multileveled partnerships (Fraser and Kenney 2000). Conflicts surrounding urban forests involve emotional, moral, and territorial dimensions (Kirkpatrick et al. 2013).

Engaging residents in decision-making processes and addressing conflicting views regarding risk, danger, or nuisance may shift negative perceptions about trees. For example, by working collaboratively with urban homeowners for the Residential Trees Survey program in Seattle (Washington), managers engaged with communities directly, understood their perspectives, and incorporated resident input into urban forest management (Dilley and Wolf 2013). Tools like Volunteer Geographic Information (VGI) could enhance participatory planning in urban forests (Foster et al. 2017). Despite VGI's opportunities for collaborations between urban forestry managers and residents in mapping trees, technologies such as VGI can result in uneven community participation and misrepresent data about who engages with this tool, emphasizing the need for careful data consideration and inclusivity in planning processes.

OPPORTUNITIES FOR FUTURE RESEARCH

Our systematic literature review highlighted challenges, contexts, and factors associated with conflicts and collaboration with urban forest management. The analysis also revealed numerous opportunities for future research to better understand conflict and collaboration regarding (1) stakeholder involvement and public education; (2) governance, policies, and decision making; (3) collaboration; and (4) biophysical assessments and human interactions and conflicts.

Stakeholder involvement and public education about urban forest management could shape the effectiveness of urban forest management. More research is needed to identify stakeholder needs in urban forestry, including more multidisciplinary research on pressures and threats related to human activities in urban and suburban forests (Živojinović and Wolfslehner 2015). It is important to understand the most effective ways to educate the public and provide stakeholder involvement opportunities (Moskell and Allred 2013; Wilkes-Allemann et al. 2015b; Davies et al. 2018; Day and Patel 2021). Although the public impacts policies, there is limited understanding of how the public can impact complex urban politics and forest management (Ryu and Chi 2021). More research is needed on recognitional equity, or the ability of practitioners and urban residents to influence governance and decision making (Nesbitt et al. 2019). Although professionals are pivotal to urban forest management, there is a need to study how knowledge is transferred between stakeholders in the fields of urban forestry and green infrastructure (Ugolini et al. 2015). There is limited research about the perspectives of professionals (Kirkpatrick et al. 2013) and the roles, development, and impacts of intermediaries on urban forest policy and management (Frantzeskaki and Bush 2021).

Stewardship activities present an important way to engage with the public and for the public to interact with urban trees. However, there was limited research about whether legal training programs impact the efficacy of volunteer-led urban forest tree committees (Harper et al. 2018), including whether stewardship on public or private lands impacts local environmental outcomes (Romolini et al. 2013). In addition, although there is increasing research about resident perceptions of urban forests and urban trees, including perceptions about conflicts, studies were usually missing details about the types of recreational participants ('passing through groups') or perspectives on forest pest management and strategies in urban forests (Gutsch et al. 2019).

Tree planting programs are often promoted as an effective way to increase urban tree canopy and reduce inequities in urban greenspaces. However, only a handful of research focuses on people's acceptance of tree giveaway programs and aftermaths, including the dynamic between urban greenspaces like urban forests and gentrification (Garrison 2019). More research is needed to assess residents' perceptions of urban trees, including their attitudes and behaviors (Tyrväinen et al. 2003; Nguyen et al. 2017), local ecological and human/social conditions and how this interacts with tree planting (Pincetl et al. 2013), and strategic governance of large-scale urban afforestation (Yao et al. 2019). In addition, resistance to street tree planting programs is not well understood, particularly for younger adults (Carmichael et al. 2019), including the behaviors and attitudes of renters, multi-family homeowners, and commercial property owners about large-scale tree plantings (Dilley and Wolf 2013), and differences in beliefs about responsibility and tree stewardship (Moskell and Allred 2013). A better understanding of public perceptions/awareness of preserved urban forests is needed as well as how to install, manage, and protect such green spaces (Jankovska et al. 2010). Urban forests serve a variety of people. However, there is little understanding of differing views on natural areas and challenges pertaining to development, preferences, activities, and current/future social conflicts (Aasetre et al. 2016). Scholars suggest using more qualitative, nonlinear, and non-written approaches to interact with people and collect data about urban trees, including key informant interviews, focus groups, stakeholder mapping, vision galleries, and sacred place mapping (Elmendorf and Luloff 2021).

Conflicts in urban forest management often result from different approaches to governance and decision making. There is a need to assess how tree canopy cover is affected by various coordination processes, how municipalities prioritize management (Ordóñez et al. 2020a), and how governance structures and policies affect the management and planning of urban forests (Wilkes-Allemann et al. 2015a, 2015b).

Policy impacts the funding, outcomes, actors, and directions of urban forestry. However, there are limited assessments about how science and data influence the planning, funding initiatives, and outcomes of urban forestry and public-private partnerships (Young 2011), along with assessments of more efficient means of integrating ecological research into the decision-making process (Driscoll et al. 2012). Furthermore, future research can assess the difference in capacities of municipal managers, including their operations (budgets, personnel, etc.), management processes, public engagement, and governance frameworks, and how these in turn impact conflicts and urban forest management, especially beyond the continental US (Ordóñez et al. 2019). Future research can also assess whether social media can influence local policies on urban forest management (Gillespie and Nguyen 2019), and which policies are most effective at preserving urban forests (Wu et al. 2019). In addition, although land use impacts urban conservation outcomes, including potential conflicts and collaboration efforts, there is limited understanding about the interplay between land-use policies, climate actions, and urban forest management, impact, outcomes, and planning processes (Cheng et al. 2021).

Owing to the diversity of stakeholders, uses, and policies involved in urban forestry, collaboration is usually a key aspect of its effective management and stewardship. On the other hand, ineffective collaboration efforts can also lead to conflicts. Therefore, there is a need to improve understanding about the success and failures of the participatory governance and urban forestry (Kozová et al. 2018), including investigations about how representative participatory and collaborative planning is (Sipilä and Tyrväinen 2005), and how the theory of social networks impacts collaborations of urban forest management (Yamaki 2016) and public-private partnerships that may impact responses to complex disturbances (Svendsen et al. 2021).

Forest management, ecological conditions, and design can impact users' perceptions, experiences, and interactions within space. Conflicts and/or collaboration may occur based on biophysical conditions. Therefore, more research is needed to assess urban forests' carrying capacity, including biophysical and social carrying capacity (Wilkes-Allemann et al. 2015b); research on tree growth/longevity; production and sales (Leibowitz 2012); and resulting impacts on conflicts and/or collaboration. Future research could further elucidate whether biological effects of different trail types/activities, recreational use, and trail-based fragmentation may impact surrounding ecosystems and potential conflicts (Ballantyne et al. 2014). In addition, more research is needed to assess how landscape design and management can better serve local ecosystems and urban residents. Invasive species management in urban areas may also be contentious for some stakeholders. However, there is limited research about invasive species management in urban ecosystems and whether it instigates stewardship and collaboration activities around urban forest management (McMillen et al. 2019). Human activities in urban ecosystems also have ripple effects on urban forests. More research is needed to assess what percent of tree damage is caused by lawnmowers (Morgenroth et al. 2015). Communication about urban forest management and conditions may also impact the development or dissolution of conflicts.

CONCLUSION

Overall, much remains to be studied about conflict and collaboration in urban forest management. As urban population increases globally, a better understanding of conflict and collaboration is critical for effective urban forest management to enhance human wellbeing and the provisioning of ecosystem services, as well as the functioning of urban forest ecosystems. The majority of the research that is available comes from more developed or industrialized countries, although urbanization is a global phenomenon. In several cases, across geographies, local contexts were highlighted as influencing or being impacted by different types of conflict or collaborations. While our systematic review highlights factors that enhance collaboration and reduce conflicts in urban forest management, our review also acknowledges that there are multiple opportunities for future research to improve collaborative decision making for the sustainability of urban forests.

LITERATURE CITED

Aasetre J, Gundersen V, Vistad OI, Holtrop EJ. 2016. Recreational preferences along a naturalness-development continuum: Results from surveys in two unequal urban forests in Europe. *Journal of Outdoor Recreation and Tourism*. 16:58-68. https://doi.org/10.1016/j.jort.2016.09.006

Alexander K, Truslove M, Davis R, Stephens S, Zentz R. 2020. A collaborative approach to preparing for and reacting to emerald ash borer: A case study from Colorado. *Forestry*. 93(2):239-253. https://doi.org/10.1093/forestry/cpz070

Ballantyne M, Gudes O, Pickering CM. 2014. Recreational trails are an important cause of fragmentation in endangered urban forests: A case-study from Australia. *Landscape and*

Urban Planning. 130:112-124. https://doi.org/10.1016/j.landurbplan.2014.07.004
Bierzychudek P. 2020. Passive recovery of an urban forest in the Pacific Northwest after removal of invasive plants. *Urban Ecosystems*. 23:1023-1038. https://doi.org/10.1007/s11252-020-00966-8
Bonilla-Duarte S, González CC, Rodríguez LC, Jáuregui-Haza UJ, García-García A. 2021. Contribution of urban forests to the ecosystem service of air quality in the city of Santo Domingo, Dominican Republic. *Forests*. 12(9):1249. https://doi.org/10.3390/f12091249
Breuste JH. 2013. Investigations of the urban street tree forest of Mendoza, Argentina. *Urban Ecosystems*. 16:801-818. https://doi.org/10.1007/s11252-012-0255-2
Brown H, Proust K, Newell B, Spickett J, Capon T, Bartholomew L. 2018. Cool communities—Urban density, trees, and health. *International Journal of Environmental Research and Public Health*. 15(7):1547. https://doi.org/10.3390/ijerph15071547
Cadaval S, Clarke M, Roman LA, Conway TM, Koeser AK, Eisenman TS. 2024. Managing urban trees through storms in three United States cities. *Landscape and Urban Planning*. 248:105102. https://doi.org/10.1016/j.landurbplan.2024.105102
Carmichael CE, McDonough MH. 2018. The trouble with trees? Social and political dynamics of street tree-planting efforts in Detroit, Michigan, USA. *Urban Forestry & Urban Greening*. 31:221-229. https://doi.org/10.1016/j.ufug.2018.03.009
Carmichael CE, McDonough MH. 2019. Community stories: Explaining resistance to street tree-planting programs in Detroit, Michigan, USA. *Society & Natural Resources*. 32(5): 588-605. https://doi.org/10.1080/08941920.2018.1550229
Cheng Z, Nitoslawski S, Konijnendijk van den Bosch CC, Sheppard S, Nesbitt L, Girling C. 2021. Alignment of municipal climate change and urban forestry policies: A Canadian perspective. *Environmental Science & Policy*. 122:14-24. https://doi.org/10.1016/j.envsci.2021.04.005
Clark C, Ordóñez C, Livesley SJ. 2020. Private tree removal, public loss: Valuing and enforcing existing tree protection mechanisms is the key to retaining urban trees on private land. *Landscape and Urban Planning*. 203:103899. https://doi.org/10.1016/j.landurbplan.2020.103899
Clark JR, Matheny NP. 1998. A model of urban forest sustainability: Application to cities in the United States. *Journal of Arboriculture*. 24(2):112-120. https://doi.org/10.48044/jauf.1998.014
Conway TM, Bang E. 2014. Willing partners? Residential support for municipal urban forestry policies. *Urban Forestry & Urban Greening*. 13(2):234-243. https://doi.org/10.1016/j.ufug.2014.02.003
Davies HJ, Doick KJ, Hudson MD, Schaafsma M, Schreckenberg K, Valatin G. 2018. Business attitudes towards funding ecosystem services provided by urban forests. *Ecosystem Services*. 32:159-169. https://doi.org/10.1016/j.ecoser.2018.07.006
Day K, Patel Z. 2021. Speaking for the trees: A study of the relationship between discourse, power and organisational culture in competing constructions of nature. *Social Dynamics*. 47(1): 172-192. https://doi.org/10.1080/02533952.2021.1904493
De Smet A, Van Reusel H. 2018. How one tree can change the future of a neighbourhood: The process behind the creation of the Boerenhof Park as an example for tactical urban planning. *Urban Forestry & Urban Greening*. 30:286-294. https://doi.org/10.1016/j.ufug.2017.09.001
Devisscher T, Ordóñez-Barona C, Dobbs C, Baptista MD, Navarro NM, Aguilar LAO, Perez JFC, Mancebo YR, Escobedo FJ. 2022. Urban forest management and governance in Latin America and the Caribbean: A baseline study of stakeholder views. *Urban Forestry & Urban Greening*. 67:127441. https://doi.org/10.1016/j.ufug.2021.127441
Dilley J, Wolf KL. 2013. Homeowner interactions with residential trees in urban areas. *Arboriculture & Urban Forestry*. 39(6): 267-277. https://doi.org/10.48044/jauf.2013.034
Doucet TC, Duinker PN, Zurba M, Steenberg JW, Charles JD. 2024. Perspectives of successes and challenges in collaborations between non-governmental organization and local government on urban forest management. *Urban Forestry & Urban Greening*. 93:128220. https://doi.org/10.1016/j.ufug.2024.128220
Driscoll CT, Lambert KF, Chapin III FS, Nowak DJ, Spies TA, Swanson FJ, Kittredge DB, Hart CM. 2012. Science and society: The role of long-term studies in environmental stewardship. *BioScience*. 62(4):354-366. https://doi.org/10.1525/bio.2012.62.4.7
Dupéy LN, Dettenmaier M, Kuhns MR, McAvoy D, Brunson MW. 2021. Paving the way: A plan for tackling urban forestry challenges and gaining public support. *Journal of Extension*. 57(6):3. https://doi.org/10.34068/joe.57.06.03
Elmendorf WF, Luloff AE. 2001. Using qualitative data collection methods when planning for community forests. *Journal of Arboriculture*. 27(3):139-151. https://doi.org/10.48044/jauf.2001.016
Fernandes CO, da Silva IM, Teixeira CP, Costa L. 2019. Between tree lovers and tree haters. Drivers of public perception regarding street trees and its implications on the urban green infrastructure planning. *Urban Forestry & Urban Greening*. 37:97-108. https://doi.org/10.1016/j.ufug.2018.03.014
Foo K. 2018. Examining the Role of NGOs in urban environmental governance. *Cities*. *77*:67-72. https://doi.org/10.1016/j.cities.2018.01.002
Foster A, Dunham IM, Kaylor C. 2017. Citizen science for urban forest management? Predicting the data density and richness of urban forest volunteered geographic information. *Urban Science*. 1(3):30. https://doi.org/10.3390/urbansci1030030
Frantzeskaki N, Bush J. 2021. Governance of nature-based solutions through intermediaries for urban transitions—A case study from Melbourne, Australia. *Urban Forestry & Urban Greening*. 64:127262. https://doi.org/10.1016/j.ufug.2021.127262
Fraser ED, Kenney WA. 2000. Cultural background and landscape history as factors affecting perceptions of the urban forest. *Journal of Arboriculture*. 26(2):106-113. https://doi.org/10.48044/jauf.2000.013
Garrison JD. 2019. Seeing the park for the trees: New York's "Million Trees" campaign vs. the deep roots of environmental inequality. *Environment and Planning B: Urban Analytics and City Science*. 46(5):914-930. https://doi.org/10.1177/2399808317737071

Gasson PE, Cutler DF. 1998. Can we live with trees in our towns and cities? *Arboricultural Journal*. 22(1):1-9. https://doi.org/10.1080/03071375.1998.9747188

Gasul D, Shmueli DF. 2016. Establishing a "community forest": Insights from the collaborative process in Migdal HaEmek, Israel. *Negotiation and Conflict Management Research*. 9(3):220-236. https://doi.org/10.1111/ncmr.12075

Gillespie J, Nguyen QH. 2019. Between authoritarian governance and urban citizenship: Tree-felling protests in Hanoi. *Urban Studies*. 56(5):977-991. https://doi.org/10.1177/0042098018784865

Gundersen V, Frivold LH, Myking T, Øyen BH. 2006. Management of urban recreational woodlands: The case of Norway. *Urban Forestry & Urban Greening*. 5(2):73-82. https://doi.org/10.1016/j.ufug.2006.06.003

Gutsch M, Larondelle N, Haase D. 2019. Of bugs and men: How forest pests and their management strategies are perceived by visitors of an urban forest. *Urban Forestry & Urban Greening*. 41:248-254. https://doi.org/10.1016/j.ufug.2019.03.003

Harper RW, Huff ES, Bloniarz DV, DeStefano S, Nicolson CR. 2018. Exploring the characteristics of successful volunteer-led urban forest tree committees in Massachusetts. *Urban Forestry & Urban Greening*. 34:311-317. https://doi.org/10.1016/j.ufug.2018.07.006

Hilbert DR, Koeser AK, Andreu M, Clarke M, Hansen G, Roman LA, Thetford M. 2023. Expanding urban tree species diversity in Florida (USA): Challenges and opportunities for practitioners. *Society & Natural Resources*. 36(8):891-908. https://doi.org/10.1080/08941920.2023.2175285

James P, Maslin M, Pringle S, Van Clief B. 2009. University green: How urban forestry partnerships plant more than trees. *Metropolitan Universities*. 20(4):75-86. https://journals.indianapolis.iu.edu/index.php/muj/article/view/20410

Jankovska I, Straupe I, Panagopoulos T. 2010. Professionals awareness in promotion of conservation and management of urban forests as green infrastructure of Riga, Latvia. *WSEAS Transactions of Environment and Development*. 6(8):614-623. http://www.wseas.us/e-library/transactions/environment/2010/88-205.pdf

Jorgensen A, Anthopoulou A. 2007. Enjoyment and fear in urban woodlands—Does age make a difference? *Urban Forestry & Urban Greening*. 6(4):267-278. https://doi.org/10.1016/j.ufug.2007.05.004

Kirkpatrick JB, Davison A, Harwood A. 2013. How tree professionals perceive trees and conflicts about trees in Australia's urban forest. *Landscape and Urban Planning*. 119:124-130. https://doi.org/10.1016/j.landurbplan.2013.07.009

Kjær AM. 2004. *Governance*. Cambridge and Malden (MA, USA): Polity Press. 240 p.

Konijnendijk van den Bosch CC. 2015. From government to governance: Contribution to the political ecology of urban forestry. In: Anders Sandberg L, Bardekjian A, Butt S, editors. *Urban forests, trees, and greenspace*. Abingdon-on-Thames (United Kingdom): Routledge. p. 35-46.

Köse M, Kirca S, Gürbey AP, Saglam S, Cinar HS, Özdemir E, Kul AA, Altincekic TH, Gümüs C. 2021. A model for urban forest management planning: Istanbul case study. *Environmental Engineering & Management Journal*. 20(7):1173-1182. http://89.44.47.69/index.php/EEMJ/article/view/4360

Kozová M, Dobšinská Z, Pauditšová E, Tomčíková I, Rakytová I. 2018. Network and participatory governance in urban forestry: An assessment of examples from selected Slovakian cities. *Forest Policy and Economics*. 89:31-41. https://doi.org/10.1016/j.forpol.2016.09.016

Kraxner F, Aoki K, Kindermann G, Leduc S, Albrecht F, Liu J, Yamagata Y. 2016. Bioenergy and the city—What can urban forests contribute? *Applied Energy*. 165:990-1003. https://doi.org/10.1016/j.apenergy.2015.12.121

Kronenberg J. 2015. Why not to green a city? Institutional barriers to preserving urban ecosystem services. *Ecosystem Services*. 12:218-227. https://doi.org/10.1016/j.ecoser.2014.07.002

Kuchelmeister G. 2000. Trees for the urban millennium: Urban forestry update. *UNASYLVA*. 51(200):49-55. https://www.fao.org/4/X3989e/x3989e09.htm

Kuhns MR, Bragg HA, Blahna DJ. 2002. Involvement of women and minorities in the urban forestry profession. *Journal of Arboriculture*. 28(1):27-34. https://doi.org/10.48044/jauf.2002.004

Lawrence A, Ambrose□Oji B. 2015. Beauty, friends, power, money: Navigating the impacts of community woodlands. *The Geographical Journal*. 181(3):268-279. https://doi.org/10.1111/geoj.12094

Lawrence A, De Vreese R, Johnston M, Konijnendijk van den Bosch CC, Sanesi G. 2013. Urban forest governance: Towards a framework for comparing approaches. *Urban Forestry & Urban Greening*. 12(4):464-473. https://doi.org/10.1016/j.ufug.2013.05.002

Leibowitz R. 2012. Urban tree growth and longevity: An international meeting and research symposium white paper. *Arboriculture & Urban Forestry*. 38(5):237-241. https://doi.org/10.48044/jauf.2012.032

Lopes AV, Farias JS. 2020. How can governance support collaborative innovation in the public sector? A systematic review of the literature. *International Review of Administrative Sciences*. 88(1):114-130. https://doi.org/10.1177/0020852319893444

Lumivero. 2020. NVivo [software]. Release 1.0. Denver (CO, USA): Lumivero. https://lumivero.com/products/nvivo

Madureira H, Monteiro A. 2021. Going green and going dense: A systematic review of compatibilities and conflicts in urban research. *Sustainability*. 13(19):10643. https://doi.org/10.3390/su131910643

Maruna M, Crnčević T, Milojević MP. 2019. The institutional structure of land use planning for urban forest protection in the post-socialist transition environment: Serbian experiences. *Forests*. 10(7):560. https://doi.org/10.3390/f10070560

Mattila O, Häyrinen L, Tervo M, Toppinen A, Berghäll S. 2015. Challenges of municipal greening and multifunctional forest management: The case of Finland. *Urban Forestry & Urban Greening*. 14(4):982-990. https://doi.org/10.1016/j.ufug.2015.09.007

McGrath D, Plummer R, Bowen A. 2021. Cultivating our urban forest future: A value-chain perspective. *FACETS*. 6(1):2084-2109. https://doi.org/10.1139/facets-2021-0076

McMillen H, Campbell LK, Svendsen ES. 2019. Weighing values and risks of beloved invasive species: The case of the survivor tree and conflict management in urban green infrastructure. *Urban Forestry & Urban Greening*. 40:44-52. https://doi.org/10.1016/j.ufug.2018.06.023

McPherson EG. 2003. Urban forestry: The final frontier? *Journal of Forestry.* 101(3):20-25. https://doi.org/10.1093/jof/101.3.20

Moher D, Liberati A, Tetzlaff J, Altman DG. 2009. Preferred reporting items for systematic reviews and meta-analyses: The PRISMA statement. *PLoS Medicine*. 6(7):e1000097. https://doi.org/10.1371/journal.pmed.1000097

Morgenroth J, Santos B, Cadwallader B. 2015. Conflicts between landscape trees and lawn maintenance equipment—The first look at an urban epidemic. *Urban Forestry & Urban Greening*. 14(4):1054-1058. https://doi.org/10.1016/j.ufug.2015.10.002

Moskell C, Allred SB. 2013. Residents' beliefs about responsibility for the stewardship of park trees and street trees in New York City. *Landscape and Urban Planning*. 120:85-95. https://doi.org/10.1016/j.landurbplan.2013.08.002

Muller RN, Bornstein C. 2010. Maintaining the diversity of California's Municipal Forests. *Arboriculture & Urban Forestry.* 36(1):18-27. https://doi.org/10.48044/jauf.2010.003

Nannini DK, Sommer R, Meyers LS. 1998. Resident involvement in inspecting trees for Dutch elm disease. *Journal of Arboriculture*. 24(1):42-46. https://doi.org/10.48044/jauf.1998.006

Nesbitt L, Meitner MJ, Girling C, Sheppard SR. 2019. Urban green equity on the ground: Practice-based models of urban green equity in three multicultural cities. *Urban Forestry & Urban Greening*. 44:126433. https://doi.org/10.1016/j.ufug.2019.126433

Nguyen VD, Roman LA, Locke DH, Mincey SK, Sanders JR, Fichman ES, Duran-Mitchell M, Tobing SL. 2017. Branching out to residential lands: Missions and strategies of five tree distribution programs in the US. *Urban Forestry & Urban Greening*. 22:24-35. https://doi.org/10.1016/j.ufug.2017.01.007

Nordström EM, Eriksson LO, Öhman K. 2010. Integrating multiple criteria decision analysis in participatory forest planning: Experience from a case study in northern Sweden. *Forest Policy and Economics*. 12(8):562-574. https://doi.org/10.1016/j.forpol.2010.07.006

O'Brien EA. 2005. Publics* and woodlands in England: Well-being, local identity, social learning, conflict and management. *Forestry*. 78(4):321-336. https://doi.org/10.1093/forestry/cpi042

O'Callaghan AM, Skelly J. 2013. Community involvement to reduce insect threats to urban forests. *The Journal of Extension.* 51(6):33. https://archives.joe.org/joe/2013december/tt8.php

Ordóñez C, Eleuterio AA, Vasquez A, Devisscher T, Baptista MD, Dobbs C, Aguilar LAO, Meléndez-Ackerman E. 2023. Views of government and non-government actors on urban forest management and governance in ten Latin-American capital cities. *Land Use Policy.* 129:106635. https://doi.org/10.1016/j.landusepol.2023.106635

Ordóñez C, Kendal D, Threlfall CG, Hochuli DF, Davern M, Fuller RA, van der Ree R, Livesley SJ. 2020a. How urban forest managers evaluate management and governance challenges in their decision-making. *Forests*. 11(9):963. https://doi.org/10.3390/f11090963

Ordóñez C, Threlfall CG, Kendal D, Hochuli DF, Davern M, Fuller RA, van der Ree R, Livesley SJ. 2019. Urban forest governance and decision-making: A systematic review and synthesis of the perspectives of municipal managers. *Landscape and Urban Planning*. 189:166-180. https://doi.org/10.1016/j.landurbplan.2019.04.020

Ordóñez C, Threlfall CG, Livesley SJ, Kendal D, Fuller RA, Davern M, van der Ree R, Hochuli DF. 2020b. Decision-making of municipal urban forest managers through the lens of governance. *Environmental Science & Policy*. 104:136-147. https://doi.org/10.1016/j.envsci.2019.11.008

Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, Shamseer L, Tetzlaff JM, Akl EA, Brennan SE, Chou R, Glanville J, Grimshaw JM, Hróbjartsson A, Lalu MM, Li T, Loder EW, Mayo-Wilson E, McDonald S, McGuinness LA, Stewart LA, Thomas J, Tricco AC, Welch VA, Whiting P, Moher D. 2021. The PRISMA 2020 statement: An updated guideline for reporting systematic reviews. *British Medical Journal*. 372:n71. https://doi.org/10.1136/bmj.n71

Pincetl S. 2010. Implementing municipal tree planting: Los Angeles million-tree initiative. *Environmental Management.* 45:227-238. https://doi.org/10.1007/s00267-009-9412-7

Pincetl S, Gillespie T, Pataki DE, Saatchi S, Saphores JD. 2013. Urban tree planting programs, function or fashion? Los Angeles and urban tree planting campaigns. *GeoJournal*. 78:475-493. https://doi.org/10.1007/s10708-012-9446-x

Pinkerton E, Rutherford M. 2021. Evaluating British Columbia's municipally owned community forest corporations as governance structures. *Environmental Management*. 67(1):1-11. https://doi.org/10.1007/s00267-020-01384-4

Pinto M, Almeida C, Pereira AM, Silva M. 2016. Urban forest governance: FUTURE—The 100,000 Trees Project in the Porto Metropolitan Area. In: Castro P, Azeiteiro U, Bacelar-Nicolau P, Leal Filho W, Azul A, editors. *Biodiversity and education for sustainable development*. Cham (Switzerland): Springer. p. 187-202. https://doi.org/10.1007/978-3-319-32318-3_12

Reams MA, Haines TK, Renner CR, Wascom MW, Kingre H. 2005. Goals, obstacles and effective strategies of wildfire mitigation programs in the wildland–urban interface. *Forest Policy and Economics*. 7(5):818-826. https://doi.org/10.1016/j.forpol.2005.03.006

Referowska-Chodak E. 2019. Management and social problems linked to the human use of European urban and suburban forests. *Forests*. 10(11):964. https://doi.org/10.3390/f10110964

Rodriguez M, Bodini A, Escobedo FJ, Clerici N. 2021. Analyzing socio-ecological interactions through qualitative modeling: Forest conservation and implications for sustainability in the peri-urban bogota (Colombia). *Ecological Modelling*. 439: 109344. https://doi.org/10.1016/j.ecolmodel.2020.109344

Roman LA, Conway TM, Eisenmann TS, Koeser AK, Ordóñez Barona C, Locke DH, Jenerette GD, Östberg J, Vogt J. 2021. Beyond 'trees are good': Disservices, management costs, and tradeoffs in urban forestry. *Ambio.* 50(3):615-630. https://doi.org/10.1007/s13280-020-01396-8

Romolini M, Grove JM, Locke DH. 2013. Assessing and comparing relationships between urban environmental stewardship networks and land cover in Baltimore and Seattle. *Landscape* *and Urban Planning*. 120:190-207. https://doi.org/10.1016/j.landurbplan.2013.08.008

Ruliffson JA, Gobster PH, Haight RG, Homans FR. 2002. Niches in the urban forest: Organizations and their role in acquiring metropolitan open space. *Journal of Forestry*. 100(6):16-23. https://doi.org/10.1093/jof/100.6.16

Ryu J, Chi SH. 2021. Slope matters: Anti-sprawl and construction of urban nature in Yongin, South Korea. *Sustainability*. 13(22):12401. https://doi.org/10.3390/su132212401

Shapira A, Shoshany M, Nir-Goldenberg S. 2013. Combining analytical hierarchy process and agglomerative hierarchical clustering in search of expert consensus in green corridors development management. *Environmental Management*. 52:123-135. https://doi.org/10.1007/s00267-013-0064-2

Shimada LD, Johnston M. 2013. Tracing the Troubles through the trees: Conflict and peace in the urban forest of Belfast, Northern Ireland. *Journal of War & Culture Studies*. 6(1):40-57. https://doi.org/10.1179/1752627212Z.0000000004

Sipilä M, Tyrväinen L. 2005. Evaluation of collaborative urban forest planning in Helsinki, Finland. *Urban Forestry & Urban Greening*. 4(1):1-12. https://doi.org/10.1016/j.ufug.2005.06.002

Sotirov M, Blum M, Storch S, Selter A, Schraml U. 2017. Do forest policy actors learn through forward-thinking? Conflict and cooperation relating to the past, present and futures of sustainable forest management in Germany. *Forest Policy and Economics*. 85:256-268. https://doi.org/10.1016/j.forpol.2016.11.011

Steenberg JWN, Duinker PN, Nitoslawski SA. 2019. Ecosystem-based management revisited: Updating the concepts for urban forests. *Landscape and Urban Planning*. 186:24-35. https://doi.org/10.1016/j.landurbplan.2019.02.006

Stobbart M, Johnston M. 2012. A survey of urban tree management in New Zealand. *Arboriculture & Urban Forestry*. 38(6):247-254. https://doi.org/10.48044/jauf.2012.033

Svendsen ES, Campbell LK, Plitt S, Johnson ML. 2021. Open for All: How are federal and municipal land management agencies adapting to the COVID-19 pandemic alongside increased societal recognition of racial injustice. *Frontiers in Sustainable Cities*. 3:25620. https://doi.org/10.3389/frsc.2021.725620

Tyrväinen L, Silvennoinen H, Kolehmainen O. 2003. Ecological and aesthetic values in urban forest management. *Urban Forestry & Urban Greening*. 1(3):135-149. https://doi.org/10.1078/1618-8667-00014

Ugolini F, Massetti L, Sanesi G, Pearlmutter D. 2015. Knowledge transfer between stakeholders in the field of urban forestry and green infrastructure: Results of a European survey. *Land Use Policy*. 49:365-381. https://doi.org/10.1016/j.landusepol.2015.08.019

Ugolini F, Sanesi G, Steidle A, Pearlmutter D. 2018. Speaking "green": A worldwide survey on collaboration among stakeholders in urban park design and management. *Forests*. 9(8):458. https://doi.org/10.3390/f9080458

United Nations. 2018. World Urbanization Prospects 2018. New York (NY, USA): United Nations, Department of Economic and Social Affairs, Population Division. [Accessed 2024 January 12]. https://population.un.org/wup/Maps

van der Jagt APN, Lawrence A. 2018. Local government and urban forest governance: Insights from Scotland. *Scandinavian Journal of Forest Research*. 34(1):53-66. https://doi.org/10.1080/02827581.2018.1532018

Vazquez-Brust D, Piao RS, de Melo MFDS, Yaryd RT, Carvalho MM. 2020. The governance of collaboration for sustainable development: Exploring the "black box". *Journal of Cleaner Production*. 256:120260. https://doi.org/10.1016/j.jclepro.2020.120260

Watkins SL, Vogt J, Mincey SK, Fischer BC, Bergmann RA, Widney SE, Westphal LM, Sweeney S. 2018. Does collaborative tree planting between nonprofits and neighborhood groups improve neighborhood community capacity? *Cities*. 74:83-99. https://doi.org/10.1016/j.cities.2017.11.006

Wilkes-Allemann J, Pütz M, Hirschi C. 2015a. Governance of forest recreation in urban areas: Analyzing the role of stakeholders and institutions using the institutional analysis and development framework. *Environmental Policy and Governance*. 25(2):139-156. https://doi.org/10.1002/eet.1668

Wilkes-Allemann J, Pütz M, Hirschi C, Fischer C. 2015b. Conflict situations and response strategies in urban forests in Switzerland. *Scandinavian Journal of Forest Research*. 30(3):204-216. https://doi.org/10.1080/02827581.2014.1002217

Wirtz Z, Hagerman S, Hauer RJ, Konijnendijk CC. 2021. What makes urban forest governance successful?—A study among Canadian experts. *Urban Forestry & Urban Greening*. 58:126901. https://doi.org/10.1016/j.ufug.2020.126901

Wu Z, Chen R, Meadows ME, Sengupta D, Xu D. 2019. Changing urban green spaces in Shanghai: Trends, drivers and policy implications. *Land Use Policy*. 87:104080. https://doi.org/10.1016/j.landusepol.2019.104080

Yamaki K. 2016. Role of social networks in urban forest management collaboration: A case study in northern Japan. *Urban Forestry & Urban Greening*. 18:212-220. https://doi.org/10.1016/j.ufug.2016.06.009

Yao N, Konijnendijk van den Bosch CC, Yang J, Devisscher T, Wirtz Z, Jia L, Duan J, Ma L. 2019. Beijing's 50 million new urban trees: Strategic governance for large-scale urban afforestation. *Urban Forestry & Urban Greening*. 44:126392. https://doi.org/10.1016/j.ufug.2019.126392

Young RF. 2011. Planting the living city: Best practices in planning green infrastructure—Results from major U.S. cities. *Journal of the American Planning Association*. 77(4):368-381. https://doi.org/10.1080/01944363.2011.616996

Zinzani A, Curzi E. 2020. Urban regeneration, forests and socio-environmental conflicts: The case of Prati di Caprara in Bologna, Italy. *ACME*. 19(1):163-186. https://doi.org/10.14288/acme.v19i1.1883

Živojinović I, Wolfslehner B. 2015. Perceptions of urban forestry stakeholders about climate change adaptation—A Q-method application in Serbia. *Urban Forestry & Urban Greening*. 14(4):1079-1087. https://doi.org/10.1016/j.ufug.2015.10.007

ACKNOWLEDGEMENTS

This research was made possible by funding from the University of Florida and the Science and Research Committee (SRC) of the International Society of Arboriculture (ISA). The authors would also like to thank Dr. Lara A. Roman, Dr. Andrew K. Koeser, and Dr. Theodore Eisenman for providing feedback on the development of this research, and Laurel Kaminsky and the University of Florida's Marston Science Library for their support in the literature search phase.

Stephanie Cadaval, PhD Candidate (corresponding author)
School of Forest, Fisheries, and Geomatic Sciences
University of Florida
PO Box 110410
Gainesville, FL, USA
+1-352-294-2739
scadaval@ufl.edu

Mysha Clarke, PhD
School of Forest, Fisheries, and Geomatic Sciences
University of Florida

Lillian Dinkins, PhD Candidate
School of Forest, Fisheries, and Geomatic Sciences
University of Florida

Ryan W. Klein, PhD
Environmental Horticulture
University of Florida

John W. Roberts, PhD
UF/IFAS Extension Palm Beach County
University of Florida

Qingyu Yang, Undergraduate Student
University of Florida

Conflicts of Interest:
The authors reported no conflicts of interest.

Résumé. La gestion des forêts urbaines est essentielle afin de promouvoir le bien-être humain, les écosystèmes et la société, notamment du fait de l'expansion de la population urbaine mondiale et de l'utilisation polyvalente de ces espaces verts urbains. Cette revue de littérature examine la conceptualisation et les facteurs qui contribuent aux conflits et/ou aux coopérations dans la gestion des forêts urbaines, incluant sans s'y limiter, les usages, les besoins et les perceptions des différents acteurs. En appliquant les méthodes PRISMA, nous avons systématiquement passé en revue 176 articles scientifiques et constaté que la plupart d'entre eux avaient été publiés entre 2013 et 2021, principalement aux États-Unis, en Australie et au Canada. Les résultats soulignent la nécessité de définir plus clairement la coopération, en mettant l'accent sur la communication, les tâches opérationnelles, la planification et les convictions partagées entre les acteurs. Les coopérations positives impliquent un engagement à plusieurs niveaux et une prise de décision inclusive. Dans la plupart des cas, de multiples enjeux ont contribué au conflit, dont une diversité de parties prenantes ayant des points de vue différents sur une situation donnée. Les conflits découlent généralement de situations complexes qui ne se prêtent pas à une solution unique et qui tendent à être une expression unique d'individus, de lieux et des points de vue impliqués. Notre revue peut renseigner les praticiens sur les pratiques de gestion davantage inclusives et adaptées au contexte des forêts urbaines. Nous concluons en présentant les enseignements obtenus et les suggestions pour de futures recherches sur l'implication des parties prenantes, l'éducation du public, la gouvernance, la politique, la prise de décision et le rôle des services biophysiques et écosystémiques dans la coopération et les conflits en matière de forêts urbaines.

Zusammenfassung. Die Bewirtschaftung städtischer Wälder ist für das menschliche Wohlbefinden, die Ökosysteme und die Gesellschaft von entscheidender Bedeutung, insbesondere angesichts der weltweit wachsenden Stadtbevölkerung und der Mehrfachnutzung dieser städtischen Grünflächen. Diese Literaturstudie untersucht die Konzeptualisierung und die Faktoren, die zu Konflikten und/oder Kooperationen in der städtischen Waldbewirtschaftung beitragen, einschließlich, aber nicht beschränkt auf die Nutzung, Bedürfnisse und Wahrnehmungen der verschiedenen Akteure. Unter Anwendung der PRISMA-Methode haben wir 176 wissenschaftliche Artikel systematisch ausgewertet und festgestellt, dass die meisten Artikel zwischen 2013 und 2021 veröffentlicht wurden und hauptsächlich aus den Vereinigten Staaten, Australien und Kanada stammen. Die Ergebnisse zeigen, dass eine klarere Definition von Zusammenarbeit erforderlich ist, die den Schwerpunkt auf Kommunikation, operative Aufgaben, Planung und gemeinsame Überzeugungen der Akteure legt. Positive Kollaborationen beinhalteten ein Engagement auf mehreren Ebenen und eine inklusive Entscheidungsfindung. In den meisten Fällen trugen mehrere Faktoren zum Konflikt bei, darunter eine Vielzahl von Interessengruppen mit unterschiedlichen Standpunkten zu einer bestimmten Situation. Bei Konflikten handelt es sich in der Regel um komplexe Situationen, die sich nicht mit einer Einheitslösung lösen lassen, sondern in der Regel eine einzigartige Manifestation der beteiligten Menschen, Orte und Perspektiven darstellen. Unsere Untersuchung kann Fachleute über integrative Praktiken und adaptives Management in städtischen Wäldern informieren. Abschließend werden Lehren und Vorschläge für die künftige Forschung in den Bereichen Einbeziehung von Interessengruppen, öffentliche Bildung, Verwaltung, Politik, Entscheidungsfindung und die Rolle biophysikalischer und ökosystemarer Dienstleistungen bei der Zusammenarbeit und bei Konflikten im städtischen Wald gegeben.

Resumen. La gestión de los bosques urbanos es crucial para apoyar el bienestar humano, los ecosistemas y la sociedad, en particular con el aumento de la población urbana mundial y los usos múltiples de estos espacios verdes urbanos. Esta revisión de la literatura examina la conceptualización y los factores que contribuyen a los conflictos y/o colaboraciones en el manejo del bosque urbano; incluidos, entre otros, los usos, necesidades y percepciones de los diversos actores. Utilizando los métodos PRISMA, revisamos sistemáticamente 176 artículos académicos y encontramos que la mayoría de los artículos se publicaron entre 2013 y 2021, principalmente de Estados Unidos, Australia y Canadá. Los hallazgos ponen de manifiesto la necesidad de definiciones más claras de colaboración, haciendo hincapié en la comunicación, las tareas operativas, la planificación y las creencias compartidas entre los actores. Las colaboraciones positivas implicaron un compromiso multinivel y una toma de decisiones inclusiva. En la mayoría de los casos, múltiples cuestiones contribuyeron al conflicto, incluida una variedad de partes interesadas con diferentes puntos de vista sobre una situación determinada. Los conflictos suelen ser situaciones complejas que no se prestan a una solución única para todos y tienden a ser una manifestación única de las personas, los lugares y las perspectivas involucradas. Nuestra revisión puede informar a los profesionales sobre prácticas más inclusivas y una gestión adaptativa en contextos de bosques urbanos. Concluimos proporcionando lecciones aprendidas y sugerencias para futuras investigaciones sobre la participación de las partes interesadas, la educación pública, la gobernanza, la política, la toma de decisiones y el papel de los servicios biofísicos y ecosistémicos en la colaboración y los conflictos de los bosques urbanos.

Arboriculture & Urban Forestry Quiz Questions

To complete this quiz, go to the ISA website, log into your MyISA account, and make your way to the page for *Arboriculture & Urban Forestry* CEU Quizzes (www.isa-arbor.com/store/ceuquizzes/113).

Add the quiz to your cart, proceed through checkout, and look for the content to appear on your personal dashboard under the header, "My Quizzes." If you need a username and password, send us an e-mail (isa@isa-arbor.com).

A passing score for this quiz requires sixteen correct answers. Quiz results will display immediately upon quiz completion. CEU(s) are processed immediately. You may take the quiz as often as is necessary to pass.